Adversarial tests of the Cancellation Atlas

Research direction · Kirtus Justus. Mathematical assistance · ChatGPT (OpenAI).

Origin of the inputs

The ten-page edition of The Cancellation Atlas identifies itself as fiction on p. 1. Page 7 explicitly grants the erasure oracle. Pages 8–9 give labels and local dependency masks. These inputs were authored as a disproof target. The external graph theorem concerns reconstruction given a cycle matroid. The supplied record contains no measured local tables or global rail outcome.

> Reject the proposed mechanism that selects space. Retain two finite results: some labelled observations reconstruct incidence under restrictive hypotheses, and a global erasure can decide whether a proposed map fits. No physical realization was supplied.

Claims under examination

Claim made by the fiction	Test and disposition
Short cancelling triples select graphic worlds	Rejected. At rank four, only 168 of 1,008 maximizing configurations are graphic. At rank five a non-graphic rival outscores every graph.
Locally reconstructible places agree globally	Rejected. Two rigid tetrahedral charts assign incompatible endpoints to a pair of shared marks. A matched triangle alone also permits extra mixed dependencies.
A single closure test certifies the specified ring	Sustained when every whole open strip is known. For three individual prisms, the rail answer alone leaves 1,519 simple candidates.
Paired ends supply physical matter and distance	Only a graph statement survives. Binary incidence gives an even endpoint set. Unit-cost hop distance is stipulated. Neither mass, propagation time nor a physical ruler is identified.

Observation convention

The columns of a binary matrix are fixed, labelled elementary marks. A closed word is a vector in its kernel. Erasing F fails when F contains a nonzero closed support; this is just column dependence on F. In a binary code each minimal dependent support specifies its word, and these words span the code. Thus complete erasure tables determine the labelled code. Throughout, “simple” for a binary representation means that its columns are distinct and nonzero. A graphic representation imposes a further restriction.

All counts refer to mathematical completions under the stated access to exact answers.

The selection rule fails

The sample space includes non-graphic column sets. The parameter β rewards cancelling triples on this entire space.

At rank four every ten-subset S of the fifteen nonzero binary vectors spans, giving 3,003 cases. A simple graph of rank r has at most r(r+1)/2 edges: a connected component of rank k has at most k(k+1)/2, and splitting positive ranks strictly decreases this bound. At rank four only K_5 attains ten. Its reduced incidence matrices form one orbit, of size |GL(4,2)|/5! = 168 by edge-labelled reconstruction [1,2].

Triples	Sets	Graphic	Marks in tetrahedral charts
8	315	0	10 / 10
9	1,680	0	10 / 10
10	1,008	168	10 / 10

An independent count takes the five omitted points. Of the 35 triples in the full space, seven pass through each point and one through each pair. Inclusion-exclusion gives $T=10-t$, where t is the number of triples wholly among the omitted points. There can be at most two. Two intersecting triples specify a five-set in $15 \times C(7,2)=315$ ways. Counting incidences leaves $35 \times C(12,2)-2\times315=1,680$ sets with one triple. The rest give the 1,008 maximizers.

Of the maximizers, 840 contain a full rank-three binary plane and three points outside it: $15\times C(8,3)=840$. The seven plane points cannot form a simple rank-three graph, whose maximum is six edges. Under weights $\exp(\beta T)$, the graphic fraction tends to 1/6, and every one of the 3,003 cases has complete local tetrahedral coverage.

Strict defeat at rank five

A fifteen-edge graph at rank five must be K_6, with 20 triangles. Use the nonzero binary vectors 1,…,15, remove 1 and 2, then add 16 and 20. Within the original hyperplane there remain $35-7-7+1=22$ triples. Outside, 16 XOR 20=4 adds another. Every mark belongs to a triple. Since $23>20$ and the ensemble is finite, the graphic probability tends to zero for these fixed parameters as β increases. This refutes the universal selection claim. It does not decide an independently specified large-rank limit.

The overlap must earn its geometry

The complete exact catalogue on a three-connected simple graph fixes that graph up to vertex relabelling, ignoring isolated vertices [1,2].

Incompatible incidence despite identical shared answers

Assign the tetrahedra vertex potentials {0,1,2,4} and {0,1,8,10}. Each edge column is the XOR of its endpoints. Their marked edge sets overlap at 1 and 2. These edges meet at vertex 0 in the first chart and are disjoint in the second. Every erasure of the two shared marks recovers in both charts. By rigidity of K_4, no relabelling resolves the incidence conflict.

A positive case and one certificate

Glue connected three-connected charts along a tree, each new chart sharing exactly a labelled triangle and its three vertices with the preceding union. Keep other vertices distinct in the proposed graph G. Let C_0 be the span of the local cycle codes extended by zeros. Then C_0 is the cycle code of G. To decompose any global even edge set, assign the shared edges to one side, producing two fragments with the same even boundary on the triangle. Add a path with that boundary to both fragments; it cancels in their binary sum. Induct along the gluing tree. The construction is a graph clique-sum [3].

Every binary completion contains C_0. Erase a spanning tree of G. If it recovers, its $|V(G)|-1$ columns are independent and the whole code has dimension at most dim C_0. Equality certifies all its erasure answers. A failure reveals an extra dependency without identifying its source. The query count excludes learning the local tables.

A limit even with many queries

A separate pair of graph realizations can have the same entire labelled code. Use two copies of the piece with terminals u,v and labelled edges uc, cv, va, ab, bu, ua; interiors in the two pieces are distinct. Glue the two terminal pairs in the same order or reversed order. An internally even edge set has equal parity at both terminals within each piece, so closing across pieces is unaffected by swapping them. The terminal degrees become (6,4) and (5,5). The graphs are nonisomorphic, yet agree on all 4,096 erasures of their twelve labels. This is a Whitney twist. A count of open endpoints can separate them only under a stronger observation contract.

Allowing parallel edges gives seven graphic laws

For the seventeen-mark pair of K_5 charts on the next page, each labelled K_5 fixes its five vertices by Whitney rigidity. The common triangle fixes the matching of its three vertices. Any further identification must match private vertices across the two pieces. There are one empty, four single and two double matchings. Their quotient kernels are distinct: a single match adds u XOR v, and each double-match plane contains precisely its two cross-piece pair vectors. Only the empty matching is simple graphic.

Seventeen marks need four queries

Two K_5 charts share the labelled triangle on vertices 0, 1 and 2. Private vertex potentials are 4,8 on the left and 16,32 on the right.

The quotient by the span of local words has dimension six. Its two local images have dimension four with a two-dimensional intersection. There are 36 one-dimensional added kernels outside both. There are 96 two-dimensional kernels disjoint from both: their projections identify the two private two-spaces in 6 ways, with 16 choices of map into the shared two-space. Including zero gives 133. Removing kernels containing a column or a pair sum leaves zero and seven lines (31,47,55,59,61,62,63). Exhaustive enumeration reproduces these counts.

The zero-kernel graph has rank six and seventeen edges. Every proper quotient has rank at most five and could have at most fifteen edges if simple graphic. The original graphic completion is therefore unique, and the two K_5 graphs glued on a triangle produce a three-connected graph. Replacing 32 by 29 imposes kernel 61. Across all 131,072 erasures this particular pair differs on 4,760, starting at size three. Erase (0,1), (4,8), (16,32): the columns 1,12,48 are independent and 1,12,13 sum to zero.

Three levels are impossible

In endpoint notation the seven proper kernels are the even sets consisting of all seven vertices except one. A forest spans precisely the sets of even parity in each of its components. The six-vertex pattern excluding v is in its span exactly when the component of v is odd and every other component is even. Hence a forest query can fail for 0,1,3,5 or 7 proper kernels; it passes on the original. A nonforest fails on all eight. No first query splits four against four as a depth-three binary decision tree would require.

Test	Erased edges
A	(0,1), (2,4), (16,32)
B	(0,1), (0,2), (4,8), (16,32)
C	(0,1), (4,8), (2,16), (2,32)
D	(0,4), (1,8), (0,16), (2,32)

In A–D order for 0,31,47,55,59,61,62,63 the signatures are 0000, 0010, 0011, 1000, 0001, 0110, 0100 and 0101. Here 1 is failure. The earlier verifier stores A as the least significant bit and prints D–A. Thus kernel 31 prints 0100 for forward 0010. Four fixed tests suffice and four are necessary even adaptively. This concerns identifying any of the eight simple binary laws. Under a prior promise of simple graphicity there is only one.

The ring needs a specified observation

m ≥ 3 layers, three sites per layer, one triangle on every layer, and three rails joining equally labelled sites in successive layers modulo m.

The earlier audit lacked the ring inputs. The later corrected audit, p. 3, names the fictional Atlas as their source. Under that source’s observation rule the three-layer ring has eighteen distinct marks: t0_01, t0_02, t0_12 and their counterparts on layers 1,2, followed by r0_0,r0_1,r0_2 and their counterparts in bands 1,2. A stipulated whole open strip omits exactly the three rails of one band. For m=3 each strip has fifteen marks and a basis of seven closed words: the three layer triangles and two rectangles for each retained band. The packet lists their supports explicitly.

Completeness of the count

For an edge word x let ∂x be its vector of odd-degree endpoints and $a_i(x)$ its rail parity in band i. Put $A(x)=(\partial x, a_{m-1}(x))$. The local codes span K=ker A: there are m independent layer triangles and 2m independent rectangles (using rails 0 and j, j=1,2). Each rectangle has a private nonzero-index rail among these generators. All lie in ker A. The ordinary incidence rank is 3m−1; the added row sees a full rail loop, so rank A=3m and dim ker A=3m. The extra row is a standard even-cycle representation [4].

Every compatible code contains K. Each open strip has rank 3m−1, since its graph is connected on all 3m sites. In V=im A its span is exactly the hyperplane $a_i=0$, by inclusion and dimension. An added subspace must avoid each of these hyperplanes except at zero. Therefore its dimension is at most one, and an added line must have every $a_i=1$. The m parity equations are independent (evaluate on one rail of each band); the complete site-0 loop solves them. This gives zero kernel plus 2^{2m} lines, or $1+4^m$ labelled laws. Every pair of marks is inspected in a strip, so simplicity loses none.

More explicitly, an added line is represented by a vector (z,1), where z has even parity on each layer. Summing the endpoints within layer i gives $a_{i-1}+a_i$, which proves the characterization. Every such z is attainable by adding layer edges to one full rail. The case z=0 gives the original cycle code. In any graph realizing the local prism tables, each labelled rigid prism fixes its rail endpoints. The matched triangles force the full rail to be a closed walk, even if extra vertex identifications are allowed. It is not closed in any nonzero-z completion or in K. Thus only z=0 is graphic. The proposed graph stays connected after removal of any two sites, so its graph realization is unique up to relabelling and invisible isolated vertices.

The price of a global answer

Two different identification questions under the whole-strip promise.

Let R contain the site-0 rail in each band. For m=3 its marks are r0_0,r1_0,r2_0 and its zero-based erasure mask is hexadecimal 9200. The graphic law fails this erasure. Every rival recovers it: R is not closed there, and each proper subset is a forest in an open strip. Any test hiding fewer than m marks also fits within an open strip and thus agrees in every law. The minimum width of a test for the proposed graph is m.

To identify any law, take T to contain triangle edges 01 and 02 on every layer. Their boundary vectors are a basis for the allowed z. Erase R∪T. Under K it recovers, since its only ordinary cycle R has odd rail parity. In each line completion there is precisely one nonzero closed support inside that set: R plus the unique subset T(z) with boundary z. Erasing R∪(T without t) recovers exactly when t occurs in T(z). These 2m further queries decode z. The whole set of queries is nonadaptive, and 2m+1 is optimal since $2^{2m}+1$ possibilities exceed the leaves of a binary decision tree of depth 2m.

Layers	Compatible laws	Narrowest graph test	Tests to identify any law
3	65	3 marks	7
4	257	4 marks	9
5	1,025	5 marks	11

For m=3 I enumerated all 262,144 erasures for the ordinary graph and an equal-rank quotient with z equal to two sites on layer 1. There are 11,205 distinguishing erasures, three of minimum size three. Every distinguishing set touches all three bands. This exhaustive comparison checks the named pair. The subspace argument, independently replayed at m=3,4,5, establishes the completion count.

Metric and physical limit

In the proposed graph, hop distance equals cyclic layer distance plus one when the within-layer labels differ. For j≥1 and m>2j+1, a radius-j ball contains 6j−1 sites. Increasing m grows a tube of fixed cross-section. A unit-cost edge, an external interval between samples, and a rule for material motion each require more structure. Neither the enumeration nor rigidity supplies these physical inputs.

Three prisms are a different experiment

For the same eighteen labels, suppose only the three individual prism tables are known. Each has nine marks, four independent closed words and rank five.

Added kernel dimension	All kernels	Simple labelled laws	Simple containing R
0	1	1	0
1	427	373	1
2	25,012	16,537	126
3	202,752	74,136	1,152
4	119,808	11,482	240
Total	348,000	102,529	1,519

The joint span of local words still has dimension nine. Each stipulated local image now has codimension four in the quotient by that span. For consistent local tables an added kernel must intersect every image trivially. To enumerate, fix the first image U and a complement D. Every kernel disjoint from U is the graph of a map from a subspace of D into U. Enumerate these maps and reject a nonzero intersection with either remaining image. For simplicity reject also a kernel containing a column or a column pair sum. The second enumeration uses reduced row-echelon form in incidence-plus-parity coordinates and agrees entry by entry.

> The rail test alone leaves 1,519 simple completions on its failure branch. Two noiseless erasure queries are necessary and sufficient to certify the graph in this three-layer example.

First erase R. Under simplicity, if its three marks fail to recover, the entire rail is closed. The code now contains the local span and R, which together generate the graph code of dimension ten. Next erase the eight-edge spanning tree made of triangle edges 01 and 02 at each layer together with r0_0 and r1_0. Recovery forces eight independent columns, hence dimension at most ten for the code, proving equality.

For the lower bound, the local span K is a strict compatible subcode. A strict compatible supercode also permits the boundary consisting of two sites on each of the three layers. Such a boundary cannot be realized in one prism or by one or two marks, preserving local data and simplicity. Anything recoverable in the graph also recovers in the subcode, and anything failing in the graph fails in the supercode. Thus no single test isolates the graph. This proof uses exactly three layers; larger prism-only networks require their own analysis.

Review record and disposition

The standalone replay packet contains the labelled inputs, executable tests and retained results.

How to reproduce the erasure claims

Labelled_Restrictions.json fixes the labels, each observed support set, and a closed-word basis for each local table. Run Replay_Local_Span.py with Python 3.10+ to reconstruct the quotient from those bases. It checks the 17-mark and 18-mark examples, all erasures for the named rival pairs, and both observation promises. Verify_Cancellation_Atlas.py checks the earlier finite controls and the general ring decoder. Verify_Four_Erasure_Queries.py counts the first-query split sizes on all original forests and checks all eight published signatures. Independent_Census.cpp provides a separate optional C++17 prism census.

Provenance of the ring result

The Cancellation Atlas, original ten-page edition of 7 November 2026, pp. 1, 7–9, SHA-256 b0001eb51c7f6490d3ca0aa157cc75ed0274767899c9460e085528e6744f37a8. The corrected Labelled Erasure Classes and Queries, p. 3, makes this origin explicit. Our completion census takes only labels and local dependency bases as input. Proposed endpoints enter a separate comparison between two named rivals. Neither calculation supplies the unobserved rail answer.

Original ten-page fictional source, stable edition. Source ↗

Corrected discrete-mathematics audit, p. 3. Source ↗

Decision

A binary dependency code can be interpreted as graph incidence in favorable cases. The preference for cancelling triples fails its own selection test. The record contains no observation selecting the graphic law. These computed cases have no evidentiary role in a quantum-readout experiment. Tests of an actual device would need independent access to named interventions, histories and matched outcomes. The fictional candidate is retired as an explanation of physical spacetime.

[1] H. Whitney (1933). 2-Isomorphic Graphs. Source ↗

[2] J. Oxley. Matroid Theory, 2nd ed., Theorem 5.3.1. Source ↗

[3] J. D. Douthitt and J. Oxley. Chordal matroids arising from generalized parallel connections. Source ↗

[4] T. Zaslavsky (1991). Biased graphs II: The three matroids. Source ↗